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Locker with valve assembly

Abstract

A locker with a valve assembly includes a locker body and an air supply assembly disposed on the locker body. A main air duct communicating with the air supply assembly is disposed inside the locker body. The locker body includes at least one storage box. The storage box includes a box body, a box door and a valve assembly. The box body includes a sidewall that surrounds a circle and further includes an inner wall connected to the rear end of the sidewall. The front end of the sidewall forms an opening. The sidewall is provided with an air vent communicating with the main air duct. The box door has a closed position for closing the opening, and an open position for opening the opening. The valve assembly includes a valve and a reset member. The valve is configured to move relative to the sidewall.

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Background/Summary

RELATED APPLICATIONS

(1) The subject application is a U.S. National Stage application of International Application No. PCT/CN2020/103821, filed on 23 Jul. 2020, which claims the priority of Chinese Patent Application No. 201910680455.5, filed on 25 Jul. 2019. The contents of which are herein incorporated by reference in their entirety.

(2) This application claims priority to Chinese Patent Application No. 201910680455.5 filed with the China National Intellectual Property Administration (CNIPA) on Jul. 25, 2019, the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(3) The present application relates to the field of item storage technology, for example, a locker with a valve assembly.

BACKGROUND

(4) To satisfy the storage requirements of fruits, vegetables, meat, aquatic products or meals that require refrigeration and heat preservation, some lockers are provided with a refrigeration assembly or a heating assembly in a locker body. However, in the actual use of these lockers, an operator may forget to close the door after taking out items. Therefore, cold air or hot air is prone to loss. As a result, energy is wasted.

SUMMARY

(5) The present application provides a locker with a valve assembly to alleviate the problem of easy loss of cold air or hot air in a case where an operator forgets to close the door of the locker.

(6) The locker with a valve assembly according to the present application includes a locker body and an air supply assembly disposed on the locker body. A main air duct communicating with the air supply assembly is disposed inside the locker body. The locker body includes at least one storage box. Each storage box includes a box body, a box door and a valve assembly. The box body includes a sidewall that surrounds a circle and further includes an inner wall connected to the rear end of the sidewall. The front end of the sidewall forms an opening. The sidewall is provided with an air vent communicating with the main air duct. The box door has a closed position for closing the opening, and an open position for opening the opening. The valve assembly includes a valve and a reset member. The valve is configured to move relative to the sidewall.

(7) When the box door is located at the closed position, the box door is configured to drive the valve to open the air vent. When the box door is located at the open position, the reset member is configured to drive the valve to close the air vent.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a structural schematic view of a locker with a valve assembly according to embodiment one of the present disclosure.

(2) FIG. 2 is a sectional structural schematic view of the locker with a valve assembly according to embodiment one of the present disclosure.

(3) FIG. 3 is a structural schematic view taken from another angle of the locker with a valve assembly according to embodiment one of the present disclosure.

(4) FIG. 4 is an enlarged partial view of the structure of part A of FIG. 3.

(5) FIG. 5 is a front view of a partial structure of a storage box of the locker with a valve assembly according to embodiment one of the present disclosure.

(6) FIG. 6 is a view illustrating the structure of a valve of the locker with a valve assembly according to embodiment one of the present disclosure.

(7) FIG. 7 is a side view of the storage box of the locker with a valve assembly when the box door of the storage box is open according to embodiment one of the present disclosure.

(8) FIG. 8 is a side view of the storage box of the locker with a valve assembly when the box door of the storage box is closed according to embodiment one of the present disclosure.

(9) FIG. 9 is a partial structural schematic view of a storage box of a locker with a valve assembly according to embodiment two of the present disclosure.

(10) FIG. 10 is a side view of a storage box of a locker with a valve assembly when the box door of the storage box is open according to embodiment three of the present disclosure.

(11) FIG. 11 is a side view of the storage box of the locker with a valve assembly when the box door of the storage box is closed according to embodiment three of the present disclosure.

REFERENCE LIST

(12) **10** locker body **20** air supply assembly **100** storage box **110** first sidewall **111** air inlet vent **112** air outlet vent **113** limiting member **114** safety lock **120** second sidewall **130** inner wall **131** second mounting hole **140** box door **150** valve **151** air inlet hole **152** air outlet hole **153** first mounting part **154** second mounting part **155** first end **156** second end **157** main body **160** reset member **170** pivot **180** first valve assembly **190** second valve assembly **158** limiting groove **310** housing **311** first plate **312** second plate **313** third plate **314** hook **320** positioning pin **200** main air duct **210** air inlet duct **220** air return duct

DETAILED DESCRIPTION

(13) Solutions of the present application are clearly and completely described below with reference to embodiments.

(14) In the description of the present application, it is to be noted that orientations or positional

relations indicated by terms such as “front”, “rear”, “up”, “down”, “left” and “right” are based on the drawings. These orientations or positional relations are intended only to facilitate and simplify the description of the present application and not to indicate or imply that a device or element referred to must have such particular orientations or must be configured or operated in such particular orientations. Thus, these orientations or positional relations are not to be construed as limiting the present application. Moreover, terms like “first” and “second” are merely for description and are not to be construed as indicating or implying relative importance.

(15) It is to be noted that in the embodiments, the direction indicated by the ab arrow in a figure represents the front-rear direction; the direction indicated by the cd arrow in the figure represents the left-right direction; and the direction indicated by the of arrow in the figure represents the up-down direction.

Embodiment One

(16) FIG. 1 is a structural schematic view of a locker with a valve assembly according to this embodiment. FIG. 2 is a sectional structural schematic view of the locker with a valve assembly according to this embodiment. FIG. 3 is a structural schematic view taken from another angle of the locker with a valve assembly according to embodiment one of the present disclosure.

(17) As shown in FIGS. 1 to 3, a locker with a valve assembly according to this embodiment includes a locker body **10** and an air supply assembly **20** disposed on the locker body **10**. A main air duct **200** communicating with the air supply assembly **20** is disposed inside the locker body **10**. The locker body **10** includes at least one storage box **100**. Each storage box **100** includes a box body, a box door **140** and a valve assembly. The box body includes a sidewall that surrounds a circle and further includes an inner wall **130** connected to the rear end of the sidewall. The front end of the sidewall forms an opening. The sidewall is provided with an air vent communicating with the main air duct **200**. The box door **140** is movably connected to the box body and has a closed position for closing the opening, and an open position for opening the opening. FIG. 4 is an enlarged partial view of the structure of part A of FIG. 3. FIG. 5 is a front view of a partial structure of a storage box of the locker with a valve assembly according to this embodiment. FIG. 6 is a perspective view illustrating the structure of a valve of the locker with a valve assembly according to this embodiment. FIG. 7 is a side view of the storage box of the locker with a valve assembly when the box door of the storage box is open according to embodiment one of the present disclosure. As shown in FIGS. 4 to 7, the valve assembly includes a valve **150** and a reset member **160**. The valve **150** is configured to move relative to the sidewall. When the box door **140** is located at the closed position, the box door **140** is configured to drive the valve **150** to open the air vent. When the box door **140** is located at the open position, the reset member **160** is configured to drive the valve **150** to close the air vent.

(18) Continuing to refer to FIG. 4, the storage box **100** further includes a safety lock **114**. The safety lock **114** is disposed between the box body and the box door **140**, and is configured to lock the box door **140** in the closed position.

(19) During the use of the locker with a valve assembly according to this embodiment, when the box door **140** of the storage box **100** is closed, the box door **140** is configured to drive the valve **150** to open the air vent. The air vent communicates with the main air duct **200**. The air supply assembly **20** is configured to send air (such as cold air or hot air) into the main air duct **200**. The air enters the box body through the air vent. The box door **140** is configured to prevent the loss of the air from the opening to an external environment. When the box door **140** of the storage box **100** is open, the reset member **160** is configured to drive the valve **150** to close the air vent to prevent the air in the main air duct **200** from entering the box body through the air vent and further losing to the external environment from the opening. In this manner, the problem of air loss in the main air duct **200** in a case where an operator forgets to close the door is effectively alleviated. Thus, energy is saved. Moreover, cost is also reduced since a requirement for an additional driving mechanism of the valve **150** is eliminated due to the force of closing the door.

(20) In this embodiment, the air supply assembly **20** may be an individual refrigeration assembly, an individual heating assembly, or an assembly with dual functions of refrigeration and heating. For example, the air supply assembly **20** may be a structure similar to an air conditioner including an evaporator, an evaporator fan, a compressor, a condenser, and a condenser fan. The compressor, the condenser and the evaporator are sequentially in a pipeline connection to form a circuit. When refrigeration is required, the evaporator fan is configured to blow the air around the evaporator into the main air duct **200**. At this time, the air is cold. The cold air may enter the inside of the box body from the main air duct **200** through the air vent to cool the items stored in the box body to achieve good storage of the items. When heating is required, the condenser fan is configured to blow the air around the condenser into the main air duct **200**. At this time, the air is hot. Similarly, the hot air may enter the inside of the box body from the main air duct **200** through the air vent to heat the items stored in the box body to achieve the good storage of the items.

(21) The refrigeration assembly or the heating assembly, or the assembly with the dual functions of refrigeration and heating is common in relevant technologies. This is not limited in the present application. As long as the cold air or the hot air can be sent to the main air duct **200**.

(22) In this embodiment, a plurality of storage boxes **100** may be divided into two columns and disposed in the locker body **10**. The two columns of storage boxes **100** are bilaterally symmetric to each other. The main air duct **200** is disposed between the two columns of storage boxes **100**. The air supply assembly **20** may simultaneously provide the cold air or the hot air to the left and right columns of storage boxes **100** through the main air duct **200** in the middle. In this manner, the volume of the locker body **10** is reduced, and the utilization of the space in the locker body **10** is improved.

(23) In this embodiment, the shape of the storage box **100** may be a cuboid, that is, the box body of the storage box **100** includes four rectangular sidewalls connected in sequence and one rectangular inner wall.

(24) The shape of the storage box **100** may also be other shapes. For example, the shape of the storage box **100** may also be cylindrical, that is, the box body of the storage box **100** includes one cylindrical sidewall and one circular inner wall. As long as a storage function can be achieved, and the disposition requirement of the valve assembly can be satisfied, the shape of the storage box **100** is not limited in the present application.

(25) In this embodiment, the sizes of a plurality of storage boxes **100** may be different. As shown in FIGS. **1** to **3**, the storage boxes **100** located in the middle have small accommodation space, and the storage boxes **100** located above and below have relatively large accommodation space. Therefore, the storage requirements of items with different volumes can be satisfied.

(26) The number, size and arrangement of the storage boxes **100** may be combined in different ways according to requirements to satisfy different storage requirements.

(27) In this embodiment, as shown in FIGS. **6** to **8**, the valve **150** is movably connected to the sidewall. The valve **150** includes a first end **155** and a second end **156** in the moving direction of the valve **150** and further includes an air hole between the first end **155** and the second end **156** and a blocking part located between the first end **155** and the second end **156**. The first end **155** is adjacent to the opening. The valve **150** has a first position for making the air hole opposite to the air vent, and a second position for making the blocking part opposite to the air vent. When the box door **140** is located at the closed position, the box door **140** is configured to abut against the first end **155** and make the valve **150** move to the first position. When the box door **140** is located at the open position, the reset member **160** is configured to drive the valve **150** to the second position. In this embodiment, the reset member **160** includes an elastic member. The elastic member is adjacent to the second end **156**. One end of the elastic member is connected to the valve **150**. Another end of the elastic member is connected to the inner wall **130**. The elastic member is configured to make the valve **150** always have a tendency to move to the second position. With this configuration, the valve assembly is simple in structure and low in cost.

(28) In this embodiment, continuing to refer to FIG. 6, the valve **150** includes a main body **157**, a first mounting part **153** disposed at the first end **155**, and a second mounting part **154** disposed at the second end **156**. The air hole is disposed on the main body **157**. The part of the main body **157** where the air hole is not disposed forms the blocking part. The box body further includes a limiting member **113** fixedly connected with the sidewall. The first mounting part **153** is connected to the limiting member **113** in an inserted manner. The second mounting part **154** is connected to the inner wall **130** in an inserted manner. When the box door **140** is located at the closed position, the first mounting part **153** is configured to abut against the box door **140**. When the box door **140** is located at the open position, the elastic member is configured to drive the valve **150** to move towards the opening. This configuration makes it possible to avoid damage to the valve **150** due to the collision between the items, and the first end **155** of the valve **150** or the second end **156** of the valve **150**, during the placement of the items into the storage box **100**.

(29) As shown in FIG. 4, the limiting member **113** is provided with a first mounting hole. The inner wall **130** is provided with a second mounting hole **131**. The first mounting hole and the second mounting hole **131** are disposed opposite each other. The first mounting part **153** is inserted into the first mounting hole. The second mounting part **154** is inserted into the second mounting hole **131**. Whether the valve **150** is located in the first position or the second position, the first mounting part **153** is always inserted into the first mounting hole, and the second mounting part **154** is always inserted into the second mounting hole **131**. Thus, it is ensured that the valve **150** can be smoothly and stably switched between the first position and the second position.

(30) In this embodiment, continuing to refer to FIG. 7, the second mounting part **154** is a strip-shaped structure that extends in a direction perpendicular to the inner wall **130**. The elastic member is a compression spring. The compression spring sleeves on the second mounting part **154**. One end of the compression spring and another end of the compression spring are configured to abut against the main body **157** and the inner wall **130** respectively. This configuration can make the structure of the valve assembly simple and compact.

(31) In this embodiment, as shown in FIG. 5, the sidewall includes a first sidewall **110** and a second sidewall **120** disposed opposite to each other in the width direction of the box body (that is, the left-right direction in FIG. 5). The first sidewall **110** is adjacent to the main air duct **200**. The air vent is disposed on the first sidewall **110**. The valve **150** is movably connected to the first sidewall **110**. As shown in FIGS. 4 and 8, the box door **140** is pivoted to the box body through a pivot **170**. The pivot **170** is adjacent to the second sidewall **120**. When the box door **140** is located at the closed position, the end of the box door **140** far away from the pivot **170** is configured to abut against the first end **155**. Since the speed of the end of the box door **140** far away from the pivot **170** is the highest during the opening of the box door **140**, with this configuration, the valve **150** can close the air vent more sensitively once the box door **140** is opened. Thus, the reliability of the valve **150** is improved. Moreover, under the action of the reset member **160**, the valve **150** is configured to abut against the box door **140** outwards and assist in driving the box door **140** to open quickly. In this manner, user's experience is also improved.

(32) The sidewall further includes a third sidewall and a fourth sidewall disposed opposite to each other in the height direction of the box body (that is, the up-down direction in FIG. 5). The first sidewall **110**, the third sidewall, the second sidewall **120** and the fourth sidewall are connected in sequence. In other embodiments, a valve **150** may be movably connected to any one of the second sidewall **120**, the third sidewall or the fourth sidewall.

(33) In other embodiments, the pivot **170** may also be adjacent to the first sidewall, the third sidewall or the fourth sidewall. Alternatively, the pivot **170** may also be directly disposed on a sidewall to make the box door **140** pivoted to a box body. In other embodiments, the box door **140** may also be connected to the box body in a slidable manner. This is not limited in the present application.

(34) In this embodiment, as shown in FIG. 2, the main air duct **200** includes an air inlet duct **210**

and an air return duct **220**. The air inlet duct **210** and the air return duct **220** extend in the up-down direction and are arranged in the front-rear direction. As shown in FIG. **4**, the air vent on the sidewall includes an air inlet vent **111** and an air outlet vent **112**. The air inlet vent **111** communicates with the air inlet duct **210**. The air outlet vent **112** communicates with the air return duct **220**. As shown in FIG. **6**, the air hole includes an air inlet hole **151** and an air outlet hole **152**. When the box door **140** is located at the closed position, the air inlet hole **151** is opposite to the air inlet vent **111**, and the air outlet hole **152** is opposite to the air outlet vent **112**. The cold air or the hot air sent by the air supply assembly **20** enters the box body through the air inlet duct **210**, the air inlet vent **111** and the air inlet hole **151**. The air in the box body returns to the air supply assembly **20** through the air outlet hole **152**, the air outlet vent **112** and the air return duct **220**. When the box door **140** is located at the open position, the air inlet vent **111** and the air outlet vent **112** are opposite to the blocking part to prevent the cold air or the hot air in the main air duct **200** from entering the box body. Therefore, the loss of the cold air or the hot air is avoided in the case where the box door **140** is open. This configuration can further simplify the structure of the valve assembly and reduce the cost.

(35) In this embodiment, a plurality of air inlet vents **111** are arranged apart in the moving direction of the valve **150**. The number of air inlet holes **151** is the same as the number of the air inlet vents **111**. When the box door **140** is located at the closed position, each of the air inlet holes **151** is opposite to a respective one of a plurality of air inlet vents **111**. A plurality of air outlet vents **112** are also arranged apart in the moving direction of the valve **150**. The number of air outlet holes **152** is the same as the number of the air outlet vents **112**. When the box door **140** is located at the closed position, each of the air outlet holes **152** is opposite to a respective one of a plurality of air outlet vents **112**.

(36) A plurality of air inlet vents **111** and a plurality of air inlet holes **151** are disposed, which increases the amount of inlet air, and accelerates the temperature change of the items in the box body, and is beneficial to maintain a good quality of the items. A plurality of air outlet vents **112** and a plurality of air outlet holes **152** are disposed, which increases the amount of return air, and air flow efficiency. Further, a guarantee is provided for maintaining the good quality of the items in the box body.

Embodiment Two

(37) FIG. **9** is a partial structural schematic view of a storage box of a locker with a valve assembly according to this embodiment. As shown in FIG. **9**, in this embodiment, a sidewall is provided with a long groove extending in a direction perpendicular to an inner wall **130**. A valve **150** is disposed in the long groove and configured to move between a first position and a second position along the long groove. The long groove is configured to support and guide the valve **150** and to make the valve **150** move smoothly between the first position and the second position along the long groove. Thus, the movement flexibility of the valve **150** is improved.

(38) The sidewall includes a housing **310**. The long groove is formed by the housing **310**. The housing **310** includes a first plate **311**, a second plate **312** and a third plate **313** vertically connected in sequence. The first plate **311** and the third plate **313** are disposed opposite to each other. The valve **150** is located between the first plate **311** and the third plate **313**. Under the joint guidance of the first plate **311** and the third plate **313**, the valve **150** can move more smoothly and stably.

(39) The second plate **312** of the housing **310** is provided with a plurality of positioning pins **320**. The valve **150** is provided with a plurality of limiting grooves **158**. A plurality of limiting grooves **158** extend in the moving direction of the valve **150**. Each of limiting grooves **158** sleeve on a respective one of a plurality of positioning pins **320**. The reset member is a compression spring located in one limiting groove **158**. One end of the compression spring is configured to abut against the inner wall of one limiting groove **158**. Another end of the compression spring is configured to abut against the positioning pin **320**. Under the action of the compression spring, the valve **150** always has a tendency to move to the first position. In this embodiment, the second plate **312** of the

housing **310** is provided with two positioning pins **320**. The valve **150** is provided with two limiting grooves **158**.

(40) When a box door **140** is located at a closed position, the box door **140** is configured to make the valve **150** move to the first position. At this time, the compression spring is compressed. An inlet hole **151** is opposite to an air inlet vent **111**. The air inlet vent **111** communicates with an air inlet duct **210**. When the box door **140** is located at an open position, under the action of an elastic restoring force of the compression spring, the valve **150** is configured to move to the second position where a blocking part is opposite to the air inlet vent **111**. Therefore, the air inlet vent **111** is closed. Moreover, when the box door **140** is open, the force exerted by the box door **140** to the valve **150** disappears. At this time, under the action of the reset member **160**, the valve **150** is reset, and the air inlet vent **111** is closed. This configuration ensures a stable state when the valve **150** is configured to move between the first position and the second position.

(41) Moreover, the problem of loss of cold air or hot air in a main air duct **200** in the case where the operator forgets to close the door is effectively alleviated. Thus, the energy is saved.

(42) The first plate **311** and the third plate **313** are each provided with a hook **314**. The opening of the hook **314** faces the inner wall **130**. Correspondingly, the sidewall is provided with a groove. The hook **314** is configured to engage with the groove in a snap-in manner. One end of the compression spring is configured to abut against the inner wall of the limiting groove **158** of the valve **150**, and another end of the compression spring is configured to abut against the positioning pin **320**. Therefore, under the action of the compression spring, the valve **150** always has a tendency to move to the second position. Under the action of the compression spring, the hook **314** of the housing **310** always maintains a tendency to engage with the groove in a snap-in manner. This configuration not only achieves a reliable guidance during the movement of the valve **150**, but also ensures the reliability of the switching of the valve **150** between the first position and the second position. Moreover, under the action of the reset member **160**, the configuration in which the housing **310** engages with the sidewall in a snap-in manner is maintained. Therefore, the mounting and the disassembly of the valve **150** can be completed without a fastener or a special tool. This configuration enables good assemblability and maintainability and simplifies the structure.

Embodiment Three

(43) FIG. **10** is a side view of a storage box of a locker with a valve assembly when the box door of the storage box is open according to this embodiment. FIG. **11** is a side view of the storage box of the locker with a valve assembly when the box door of the storage box is closed according to this embodiment. As shown in FIGS. **10** and **11**, this embodiment provides another locker with a valve assembly. The difference between this locker with a valve assembly and embodiment one is that: an air inlet vent **111** and an air outlet vent **112** are disposed apart in a vertical direction; the valve assembly includes a first valve assembly **180** and a second valve assembly **190**; when a box door **140** is located at a closed position, an air hole of the first valve assembly **180** is opposite to the air inlet vent **111**, and an air hole of the second valve assembly **190** is opposite to the air outlet vent **112**; and when the box door **140** is located at an open position, a blocking part of the first valve assembly **180** is opposite to the air inlet vent **111**, and a blocking part of the second valve assembly **190** is opposite to the air outlet vent **112**.

(44) In use, when the box door **140** is closed, the box door **140** is configured to abut against a first end **155** of a valve **150** in the first valve assembly **180**, and a first end **155** of a valve **150** in the second valve assembly **190**. Therefore, two valves **150** are driven to move towards an inner wall **130**. The air hole of the first valve assembly **180** is opposite to the upper air inlet vent **111**. The air hole of the second valve assembly **190** is opposite to the lower air outlet vent **112**. Thus, the inflow and outflow of cold air or hot air can be achieved. When the box door **140** is opened, in the first valve assembly **180**, a reset member **160** is configured to drive the valve **150** to move towards the opening, and the blocking part of the valve **150** is configured to close the air inlet vent **111**. In the

second valve assembly **190**, a reset member **160** is configured to drive the valve **150** to move towards the opening, and the blocking part of the valve **150** is configured to close the air outlet vent **112**. Thus, the loss of cold air or hot air in a main air duct **200** is prevented.

(45) In addition, the air inlet vent **111** and the air outlet vent **112** are disposed apart in the vertical direction to make the distance between the air inlet vent **111** and the air outlet vent **112** far. Thus, after the cold air or the hot air enters a box body through the air inlet vent **111** and before the cold air or the hot air flows out the box body through the air outlet vent **112**, the cold air or the hot air can more fully exchange heat with air or items in the box body. Further, a refrigeration effect or a heating effect is improved.

(46) Other features of this embodiment are the same as those of embodiment one. The technical effects of each solution have been described in embodiment one. Thus, the details are not repeated here.

Claims

1. A locker with a valve assembly, comprising a locker body and an air supply assembly disposed on the locker body, and a main air duct communicating with the air supply assembly is disposed inside the locker body; the locker body comprises at least one storage box, each of the at least one storage box comprises a box body, a box door and the valve assembly, the box body comprises a sidewall that surrounds a circle and further comprises an inner wall connected to a rear end of the sidewall, a front end of the sidewall forms an opening, and the sidewall is provided with an air vent communicating with the main air duct; the box door has a closed position for closing the opening, and an open position for opening the opening; and the valve assembly comprises a valve and a reset member, wherein the valve is configured to move relative to the sidewall; and when the box door is located at the closed position, the box door is configured to drive the valve to open the air vent; and when the box door is located at the open position, the reset member is configured to drive the valve to close the air vent; wherein the valve is movably connected to the sidewall; the valve comprises a first end and a second end in a moving direction of the valve and further comprises an air hole located between the first end of the valve and the second end of the valve and a blocking part located between the first end of the valve and the second end of the valve, and the first end of the valve is adjacent to the opening; and the valve has a first position for making the air hole opposite to the air vent, and a second position for making the blocking part opposite to the air vent; and when the box door is located at the closed position, the box door is configured to abut against the first end of the valve and make the valve move to the first position; and when the box door is located at the open position, the reset member is configured to drive the valve to the second position; wherein the reset member comprises an elastic member, wherein a first end of the elastic member is connected to the valve, a second end of the elastic member is connected to the inner wall or the sidewall, and the elastic member is configured to make the valve have a tendency to move to the second position; wherein the valve comprises a main body, a first mounting part disposed at the first end of the valve, and a second mounting part disposed at the second end of the valve, the air hole is disposed on the main body, and a part of the main body where the air hole is not disposed forms the blocking part; and the box body further comprises a limiting member fixedly connected with the sidewall, the first mounting part is connected to the limiting member in an inserted manner, and the second mounting part is connected to the inner wall in an inserted manner; and when the box door is located at the closed position, the first mounting part is configured to abut against the box door, and when the box door is located at the open position, the elastic member is configured to drive the valve to move towards the opening.

2. The locker with the valve assembly according to claim 1, wherein the second mounting part is a strip-shaped structure that extends in a direction perpendicular to the inner wall, the elastic member is a compression spring, the compression spring sleeves on the second mounting part, and one end

of the compression spring and another end of the compression spring are configured to abut against the main body and the inner wall respectively.

3. The storage cabinet with the valve assembly according to claim 2, wherein the sidewall comprises a first sidewall and a second sidewall disposed opposite to each other in a width direction of the box body, wherein the first sidewall is adjacent to the main air duct, the air vent is disposed on the first sidewall, and the valve is movably connected to the first sidewall; and the box door is pivoted to the box body through a pivot, the pivot is adjacent to the second sidewall; and when the box door is located at the closed position, an end of the box door facing away from the pivot is configured to abut against the first end of the valve.

4. The storage cabinet with the valve assembly according to claim 2, wherein the main air duct comprises an air inlet duct and an air return duct, and the air vent comprises an air inlet vent and an air outlet vent, wherein the air inlet vent communicates with the air inlet duct, and the air outlet vent communicates with the air return duct; and the air hole comprises an air inlet hole and an air outlet hole, wherein when the box door is located at the closed position, the air inlet hole is opposite to the air inlet vent, and the air outlet hole is opposite to the air outlet vent; and when the box door is located at the open position, the air inlet vent and the air outlet vent are each opposite to the blocking part.

5. The storage cabinet with the valve assembly according to claim 2, wherein the sidewall is provided with a long groove extending in a direction perpendicular to the inner wall, and the valve is disposed in the long groove and configured to move between the first position and the second position along the long groove.

6. The locker with the valve assembly according to claim 1, wherein the sidewall comprises a first sidewall and a second sidewall disposed opposite to each other in a width direction of the box body, wherein the first sidewall is adjacent to the main air duct, the air vent is disposed on the first sidewall, and the valve is movably connected to the first sidewall; and the box door is pivoted to the box body through a pivot, the pivot is adjacent to the second sidewall; and when the box door is located at the closed position, an end of the box door far away facing away from the pivot is configured to abut against the first end of the valve.

7. The locker with the valve assembly according to claim 1, wherein the main air duct comprises an air inlet duct and an air return duct, and the air vent comprises an air inlet vent and an air outlet vent, wherein the air inlet vent communicates with the air inlet duct, and the air outlet vent communicates with the air return duct; and the air hole comprises an air inlet hole and an air outlet hole, wherein when the box door is located at the closed position, the air inlet hole is opposite to the air inlet vent, and the air outlet hole is opposite to the air outlet vent; and when the box door is located at the open position, the air inlet vent and the air outlet vent are each opposite to the blocking part.

8. The locker with the valve assembly according to claim 7, wherein a plurality of air inlet vents are disposed, a plurality of air inlet holes are disposed, a number of the plurality of air inlet holes is the same as a number of the plurality of air inlet vents, and when the box door is located at the closed position, each of the air inlet holes is opposite to a respective one of the plurality of air inlet vents.

9. The storage cabinet with the valve assembly according to claim 8, wherein a plurality of air outlet vents are disposed, a plurality of air outlet holes are disposed, a number of the plurality of air outlet holes is the same as a number of the plurality of air outlet vents, and when the box door is located at the closed position, each of the air outlet holes is opposite to a respective one of the plurality of air outlet vents.

10. The locker with the valve assembly according to claim 7, wherein a plurality of air outlet vents are disposed, a plurality of air outlet holes are disposed, a number of the plurality of air outlet holes is the same as a number of the plurality of air outlet vents, and when the box door is located at the closed position, each of the air outlet holes is opposite to a respective one of the plurality of air outlet vents.

11. The locker with the valve assembly according to claim 1, wherein the main air duct comprises an air inlet duct and an air return duct, and the air vent comprises an air inlet vent and an air outlet vent disposed apart in a vertical direction, wherein the air inlet vent communicates with the air inlet duct, and the air outlet vent communicates with the air return duct; and the valve assembly comprises a first valve assembly and a second valve assembly, wherein when the box door is located at the closed position, an air hole of the first valve assembly is opposite to the air inlet vent, and an air hole of the second valve assembly is opposite to the air outlet vent; and when the box door is located at the open position, a blocking part of the first valve assembly is opposite to the air inlet vent, and a blocking part of the second valve assembly is opposite to the air outlet vent.

12. The locker with the valve assembly according to claim 1, wherein the sidewall is provided with a long groove extending in a direction perpendicular to the inner wall, and the valve is disposed in the long groove and configured to move between the first position and the second position along the long groove.

13. The locker with the valve assembly according to claim 12, wherein the sidewall comprises a housing, the long groove is formed by the housing, and the housing comprises a first plate, a second plate and a third plate vertically connected in sequence, wherein the first plate and the third plate are disposed opposite to each other, and the valve is located between the first plate and the third plate.

14. The locker with the valve assembly according to claim 13, wherein the second plate is provided with a plurality of positioning pins, and the valve is provided with a plurality of limiting grooves, wherein the plurality of limiting grooves extend in the moving direction of the valve, and each of the plurality of limiting grooves sleeves on a respective one of the plurality of positioning pins.

15. The locker with the valve assembly according to claim 14, wherein the reset member is a compression spring located in one limiting groove of the plurality of limiting grooves, wherein one end of the compression spring is configured to abut against an inner wall of the one limiting groove, another end of the compression spring is configured to abut against one positioning pin of the plurality of positioning pins sleeved on by the one limiting groove of the plurality of limiting grooves, and under an action of the compression spring, the valve has a tendency to move to the second position.

16. The locker with the valve assembly according to claim 15, wherein the first plate and the third plate are each provided with a hook, an opening of the hook faces the inner wall, the sidewall is provided with a groove, and the hook is configured to engage with the groove in a snap-in manner.

17. A locker with a valve assembly, comprising a locker body and an air supply assembly disposed on the locker body, and a main air duct communicating with the air supply assembly is disposed inside the locker body; the locker body comprises at least one storage box, each of the at least one storage box comprises a box body, a box door and the valve assembly, the box body comprises a sidewall that surrounds a circle and further comprises an inner wall connected to a rear end of the sidewall, a front end of the sidewall forms an opening, and the sidewall is provided with an air vent communicating with the main air duct; the box door has a closed position for closing the opening, and an open position for opening the opening; and the valve assembly comprises a valve and a reset member, wherein the valve is configured to move relative to the sidewall; and when the box door is located at the closed position, the box door is configured to drive the valve to open the air vent; and when the box door is located at the open position, the reset member is configured to drive the valve to close the air vent; wherein the valve is movably connected to the sidewall; the valve comprises a first end and a second end in a moving direction of the valve and further comprises an air hole located between the first end of the valve and the second end of the valve and a blocking part located between the first end of the valve and the second end of the valve, and the first end of the valve is adjacent to the opening; and the valve has a first position for making the air hole opposite to the air vent, and a second position for making the blocking part opposite to the air vent; and when the box door is located at the closed position, the box door is configured to abut against the

first end of the valve and make the valve move to the first position; and when the box door is located at the open position, the reset member is configured to drive the valve to the second position; wherein the main air duct comprises an air inlet duct and an air return duct, and the air vent comprises an air inlet vent and an air outlet vent, wherein the air inlet vent communicates with the air inlet duct, and the air outlet vent communicates with the air return duct; and the air hole comprises an air inlet hole and an air outlet hole, wherein when the box door is located at the closed position, the air inlet hole is opposite to the air inlet vent, and the air outlet hole is opposite to the air outlet vent; and when the box door is located at the open position, the air inlet vent and the air outlet vent are each opposite to the blocking part.
